Joao Coelho

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Summary

PhD candidate in a dual PhD program between Carnegie Mellon University and Instituto Superior Técnico, expecting to graduate in mid-2027. My current research focuses on LLM agents and dense retrieval, with broader interests in Machine Learning, Natural Language Processing, and Information Retrieval.

Education

Carnegie Mellon University, Pittsburgh, PA – Dual PhD in Language Technologies — Sep 2022 – Present
- Advisor: Prof. Chenyan Xiong. Collaborated with Prof. Jamie Callan.

Instituto Superior Técnico, Lisbon, Portugal – Dual PhD in Electrical Engineering — Sep 2022 – Present
- Advisors: Prof. Bruno Martins and Prof. Joao Magalhaes.

Instituto Superior Técnico, Lisbon, Portugal – MSc in Computer Science and Engineering — Sep 2021
- Merit Distinctions: 2019/2020 and 2020/2021.
- Thesis: Geographical Question Answering Leveraging Neural Language Models for Passage Retrieval.

Instituto Superior Técnico, Lisbon, Portugal – BSc in Computer Science and Engineering — Sep 2019
- Merit Distinctions: 2017/2018 and 2018/2019.

Selected Publications

Preprints

- **Joao Coelho**, Hong Wang, Jie Yuan, Zhuoer Wang, Samson Koelle, and Wei Niu. *Learning Multi-Step Query Rewriting via Corpus Feedback for Conversational Search*. arXiv (under review), 2026.
- **Joao Coelho**, Joao Magalhaes, Bruno Martins, and Chenyan Xiong. *Effective Reinforcement Learning for Agentic Search by Recycling Zero-Variance Queries During Training*. arXiv (under review), 2026.
- Sidhaarth Murali, **Joao Coelho**, Jingjie Ning, Joao Magalhaes, Bruno Martins, and Chenyan Xiong. *Beyond Parallel Sampling: Diverse Query Initialization for Agentic Search*. arXiv (under review), 2026.

International Conferences

- **Joao Coelho**, Jingjie Ning, Jingyuan He, Kangrui Mao, Abhijay Sai Paladugu, Pranav Setlur, Jiahe Jin, Jamie Callan, Joao Magalhaes, Bruno Martins, and Chenyan Xiong. *DeepResearchGym: A Free, Transparent, and Reproducible Sandbox for Deep Research*. **ICTIR 2026**.
- Jingjie Ning*, **Joao Coelho***, Yibo Kong*, Yunfan Long, Bruno Martins, Joao Magalhaes, Jamie Callan, and Chenyan Xiong. *Agentic Search in the Wild: Intents and Trajectory Dynamics from 14M+ Real Search Requests*. **SIGIR 2026**. (*equal contribution)
- **Joao Coelho**, Bruno Martins, Joao Magalhaes, and Chenyan Xiong. *Aligning Web Query Generation with Ranking Objectives via Direct Preference Optimization*. **SIGIR 2025**.
- **Joao Coelho**, Bruno Martins, Joao Magalhaes, Jamie Callan, and Chenyan Xiong. *Dwell in the Beginning: How Language Models Embed Long Documents for Dense Retrieval*. **ACL 2024**.
- **Joao Coelho**, Joao Magalhaes, and Bruno Martins. *Improving Neural Models for the Retrieval of Relevant Passages to Geographical Queries*. **SIGSPATIAL 2021**.

International Journals

- **Joao Coelho**, Beatriz Paula, Diogo Mano, Carlos Coutinho, Joao Oliveira, Ricardo Ribeiro, and Fernando Batista. *Semantic Similarity for Mobile Application Recommendation under Scarce User Data*. **Engineering Applications of Artificial Intelligence**, 2023.

Experience

Applied Scientist Intern, Amazon Web Services – Seattle, WA May 2025 – Aug 2025
Supervisors: Dr. Wei Niu and Dr. Hong Wang

- Built distributed training and inference infrastructure to support extensive dense and sparse retrieval experimentation across knowledge bases scaling up to 136 million documents.
- Proposed a search-aware multi-turn query rewriting agent, which achieved a 10% absolute hit rate improvement over the team's production solution on internal customer service search benchmarks.
- Improved efficiency by distilling the agentic behavior into a single-turn, locally deployable language model, retaining 98% of the performance while running 4 times faster.
- Submitted the work as a conference paper (under review).

Machine Learning Engineer, Caixa Mágica Software – Lisbon, Portugal Sep 2020 – Sep 2022
Supervisor: Prof. Carlos Coutinho

- Proposed a self-supervised BERT-based recommendation system for mobile applications, achieving a 10% absolute improvement in real-user satisfaction in A/B testing on a major app store.
- Led and presented a proof-of-concept integrating graph databases and neural entity extraction for the Portuguese Archive, securing one of the company's largest client projects.
- Built the machine learning pipeline that extended this proof-of-concept, enabling entity extraction and disambiguation over a large historical document collection.
- Supervised interns across recommendation and NLP projects; led technical workshops and co-authored multiple peer-reviewed papers.

Intern Developer, Caixa Mágica Software – Lisbon, Portugal Jul 2020 – Sep 2020

- Built an FAQ retrieval system for an automated helpdesk platform under low-resource conditions.
- Deployed the solution by integrating retrieval models with a React front-end using Docker and CI/CD pipelines.

Intern Developer, Xpand IT – Lisbon, Portugal Jul 2019 – Sep 2019

- Implemented a proactive chatbot with notification and polling capabilities for internal MS Teams integration.
- Integrated LUIS language understanding and deployed the production system using .NET and Azure infrastructure.

Teaching

Teaching Assistant, Carnegie Mellon University – Pittsburgh, PA
Supervisor: Prof. Chenyan Xiong

- Large Language Models: Methods and Applications (Fall 2024)

Teaching Assistant, Instituto Superior Técnico – Lisbon, Portugal
Supervisor: Prof. Ines Lynce

- Search and Planning (Fall 2021, Fall 2022)
- Artificial Intelligence (Spring 2021, Spring 2022)

Skills

Machine Learning & NLP: PyTorch, Transformers, LLM (pre-, mid-, post-)training, Knowledge Distillation, Dense Retrieval, Re-ranking, ANN, Recommendation Systems, Question Answering, RAG, LLM Agents, Search Agents
Engineering & Cloud: Distributed Training/Inference, AWS, Azure, Docker, CI/CD
Databases: SQL, NoSQL, Graph Databases, Elasticsearch
Languages: Portuguese (native), English (C1)